

Rhutika Patil

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EDUCATION

North Carolina State University

Master of Science, Bioinformatics

Aug 2024 – Aug 2026

Raleigh, NC

PCET's Nutan Maharashtra Institute of Engineering & Technology (Savitribai Phule Pune University)

Bachelor of Engineering, Electronics & Telecommunication Engineering

Aug 2018 – Jul 2022

Pune, India

SKILLS

Genomics & NGS: RNA-seq, scRNA-seq, Tag-seq, GWAS QC, FASTQ/BAM/VCF, FastQC, MultiQC, STAR, HISAT2, Salmon, Bowtie, GATK, BLAST, PLINK

Analysis: Seurat, Scanpy, SCTransform, DESeq2, edgeR, Bioconductor, differential expression, pathway analysis, cell-type annotation

Programming: Python, R, SQL, Bash; NumPy, pandas, scikit-learn, Matplotlib, ggplot2, R Markdown

ML & Statistics: Machine learning, NLP, statistical modeling, multivariate analysis, dimensionality reduction, survival analysis

Pipelines & Infrastructure: Snakemake, Apache Airflow, Docker, Linux, HPC/SLURM, Galaxy, Git/GitHub, Terraform, AWS

Data, BI & Clinical: PostgreSQL, MySQL, Snowflake, ETL/ELT, Power BI, Tableau, Excel; CDISC SDTM, ADaM, TLF; HIPAA, HL7, FHIR

WORK EXPERIENCE

Bioinformatics Consultant

North Carolina State University

Jan 2026 – Jun 2026

Raleigh, NC

- Built end-to-end RNA-seq and Tag-seq pipelines in R and Python for 50+ multi-species libraries, cutting preprocessing time by 40%.
- Sped up cross-species statistical analysis by 30% and quantified expression across 20,000+ genes to prioritize follow-up targets.
- Wrote documentation and workflow guides so teammates with varied technical backgrounds could run the pipelines independently.

Bioinformatics Research Analyst

North Carolina State University

Aug 2025 – Dec 2025

Raleigh, NC

- Integrated physiological and environmental data from 100+ field observations in R to model plant drought response.
- Modeled trait-environment relationships with multivariate statistics, improving reliability of drought-tolerance indicators by 25%.
- Identified drought-tolerance signatures with regression and ggplot2 visualizations, and presented findings to research teams.

Bioinformatics Graduate Research Assistant

NC State College of Veterinary Medicine

May 2024 – Aug 2025

Raleigh, NC

- Built RNA-seq and scRNA-seq pipelines in R and Seurat, processing ~90 GB of sequencing data from 15 samples across conditions.
- Automated QC, alignment and differential expression workflows, cutting manual processing time by 40%.
- Analyzed 10,000+ single cells for clustering and cell-type annotation; interpreted results with veterinary scientists.

Bioinformatics Graduate Research Assistant

Auburn University

Jan 2024 – Apr 2024

Auburn, AL

- Analyzed bulk RNA-seq from 6+ samples with alignment and normalization pipelines to study host responses to bacterial infection.
- Characterized differential expression patterns to identify molecular drivers of the host response to the bacterial pathogen.
- Interpreted pathway-level responses to uncover immune and inflammatory signaling mechanisms.

Bioinformatics Software Engineer

CitiusTech

Jul 2022 – Aug 2023

India

- Modeled 1M+ healthcare records in Python and R with cross-functional teams to generate population-level clinical insights.
- Built HIPAA-, HL7- and FHIR-compliant validation and ETL pipelines, improving efficiency and data integrity across clinical systems.
- Developed Power BI and Tableau dashboards combining clinical and operational metrics for department-level decisions.

PROJECTS

FAERS Drug Safety Signal Detection | Python, scikit-learn, NLP, Docker

- Recovered 17/17 known signals via PRR/ROR on 660 drug-reaction pairs; NLP on narratives reached ROC-AUC 0.85 vs. 0.76.

scRNA-seq PBMC Analysis Pipeline | Python, Scanpy, Snakemake, Docker

- Config-driven Snakemake workflow for QC, PCA, Leiden clustering, UMAP and automated cell-type annotation of 2,638 PBMCs.

Clinical Statistical Programming in R | SDTM, ADaM, TLF

- Programmed SDTM domains and ADaM datasets (ADSL, ADAE, ADTTE) with XPT export; built TLFs with Kaplan-Meier and Cox models.

Public Health Cloud ETL Pipeline | Apache Airflow, PostgreSQL, Docker, Terraform

- Validated and loaded live public-health data (61,942 → 8,899 weekly rows) with reconciliation checks and regression tests.

CERTIFICATIONS

Google Data Analytics Professional Certificate (Google) • Machine Learning Specialization (Coursera)